DATABASE TABLES & INPUT REFERENCE

CityHospital and Medical Assistant

Code review: 1 November 2026

What this document covers

This document lists the persisted database tables found in the project, the columns and constraints in each table, and the expected input or meaning of each column. Relationship descriptions are written as sentences after the tables.

Two separate databases

CityHospital currently defines one persisted table: users.
Medical Assistant uses a separate SQLite database with six tables.
Patient references do not connect the two databases in the current code.

Input notes

Inputs describe either values supplied by an API/form or values generated by the backend. IDs and timestamps are generally generated automatically. JSON values are stored as text in the current Medical Assistant database.

Tables in this document

- CityHospital - users
- Medical Assistant - conversations
- Medical Assistant - messages
- Medical Assistant - patient_records
- Medical Assistant - doctors
- Medical Assistant - appointments
- Medical Assistant - summaries

CityHospital - users

Stores all CityHospital accounts. The role value distinguishes patient, doctor, and admin accounts.

Column / key	Type / constraint	Expected input / meaning
id (PK)	INTEGER; auto-generated; indexed	Unique account ID.
full_name	VARCHAR(100); NOT NULL	Account holder name. Patient registration accepts 3-50 characters; English/Urdu letters, spaces, period, apostrophe, and hyphen.
email	VARCHAR(255); NOT NULL; UNIQUE; indexed	Valid email address, stored lowercase.
password_hash	VARCHAR(255); NOT NULL	Server-generated password hash; the submitted password itself is never stored.
role	VARCHAR(20); NOT NULL; default patient	patient, doctor, or admin. Self-registration always creates a patient.
phone	VARCHAR(20); nullable	Patient registration requires a Pakistani mobile number: 03XXXXXXXX or +923XXXXXXXX.
date_of_birth	DATE; nullable	Patient registration requires a past date no more than 120 years ago.
gender	VARCHAR(10); nullable	Patient registration accepts male, female, or other.
is_active	BOOLEAN; NOT NULL; default true	Whether the account is active and can log in.
created_at	DATETIME with timezone	Account creation time, set by the backend in UTC.

Medical Assistant - conversations

Stores a patient-reference chat session. Created when a conversation is started.

Column / key	Type / constraint	Expected input / meaning
id (PK)	TEXT	Conversation UUID generated by the backend.
patient_id	TEXT; NOT NULL	Patient reference supplied by the client; maximum 128 characters. No patient table or FK validates it.
title	TEXT; NOT NULL; default New conversation	Optional title, maximum 160 characters. The first user message replaces the default title.
state	TEXT; NOT NULL; default {}	JSON serialized conversation state, used for appointment flow.
created_at	TEXT; default current timestamp	Time the conversation was created.
updated_at	TEXT; default current timestamp	Time the conversation was last updated; used to order patient sessions.

Medical Assistant - messages

Stores each user message and assistant reply in a conversation.

Column / key	Type / constraint	Expected input / meaning
id (PK)	INTEGER; AUTOINCREMENT	Unique message row ID; increasing IDs provide message order.
conversation_id (FK)	TEXT; NOT NULL	Owning conversation ID; references conversations.id.
role	TEXT; NOT NULL	Message sender: user or assistant.
content	TEXT; NOT NULL	Plain-text message or assistant response.
created_at	TEXT; default current timestamp	Time the message was created.
structured	TEXT; nullable	Optional JSON structured medical assessment; added to existing databases through ALTER TABLE.

Medical Assistant - patient_records

Stores patient-specific source facts used by the assistant to answer questions about reports, prescriptions, and other records.

Column / key	Type / constraint	Expected input / meaning
id (PK)	INTEGER; AUTOINCREMENT	Unique patient-record row ID.
patient_id	TEXT; NOT NULL	Patient reference, maximum 128 characters; no patient table or FK.
category	TEXT; NOT NULL	Record type, maximum 60 characters (for example, report or prescription).
content	TEXT; NOT NULL	Source record text, maximum 50,000 characters.
metadata	TEXT; NOT NULL; default {}	JSON object for extra source metadata.
created_at	TEXT; default current timestamp	Time the record was added.

Medical Assistant - doctors

Stores doctor directory details and available appointment slots.

Column / key	Type / constraint	Expected input / meaning
id (PK)	TEXT	External doctor identifier, 1-100 characters.
name	TEXT; NOT NULL	Doctor name, 1-160 characters.
specialty	TEXT; NOT NULL	Medical specialty, 1-160 characters.
hospital	TEXT; NOT NULL	Hospital name, 1-200 characters.
availability	TEXT; NOT NULL; default []	JSON array of slot strings, maximum 100 items; booked slots are filtered using appointments.

Medical Assistant - appointments

Stores a doctor slot after the patient confirms the booking.

Column / key	Type / constraint	Expected input / meaning
id (PK)	TEXT	Appointment UUID generated by the backend.
patient_id	TEXT; NOT NULL	Patient reference; no patient table or FK.
doctor_id	TEXT; NOT NULL	Doctor identifier. Application uses it to match doctors, but database declares no FK.
slot	TEXT; NOT NULL	Selected available time string (example: 2026-11-02 10:00).
status	TEXT; NOT NULL; default booked	Appointment state; booked rows mark a slot as occupied.
created_at	TEXT; default current timestamp	Time the appointment row was created.
(doctor_id, slot)	UNIQUE constraint	Prevents the same doctor slot from being booked twice.

Medical Assistant - summaries

Stores structured summaries produced from uploaded medical files. The uploaded file itself is not stored.

Column / key	Type / constraint	Expected input / meaning
id (PK)	TEXT	Summary UUID generated by the backend.
patient_id	TEXT; NOT NULL; default empty	Optional patient reference supplied with upload; no patient table or FK.
fields	TEXT; NOT NULL	JSON object with patient_overview, chief_complaint, medical_history, surgical_history, medications, allergies, family_history, social_history, vitals_and_results, and assessment_and_plan.
source_name	TEXT; NOT NULL	Uploaded filename (or uploaded-record when absent).
created_at	TEXT; default current timestamp	Time the summary row was created.

Expected relationships (sentence form)

Messages and conversations

One conversation can contain zero or many messages, and each message belongs to one conversation through messages.conversation_id referencing conversations.id. This is the only declared relationship foreign key in the Medical Assistant tables.

Doctors and appointments

One doctor can have zero or many appointments by matching doctors.id to appointments.doctor_id in application logic. The database does not declare this as a foreign key.

Patient-owned assistant data

One patient reference may be associated with many conversations, patient records, appointments, and summaries. These are logical associations only because patient_id is text and no patient table or foreign key exists in the Medical Assistant database.

CityHospital users and Medical Assistant records

The CityHospital users table has no declared relationship to the Medical Assistant tables. The current assistant patient_id is an arbitrary text reference and is not linked to users.id.

Other request inputs and stored data

CityHospital registration

The request supplies full_name, email, phone, date_of_birth, gender, and password. The role is not accepted from the registering client; the backend assigns patient. Only password_hash is stored.

Medical Assistant chat

The client supplies patient_id and message text. The conversation and messages are persisted; the generated structured response is optional JSON in messages.structured.

Medical record integration

The integration supplies patient_id, category, content, and optional metadata. Metadata is a JSON object stored as text.

Doctor integration

The integration supplies id, name, specialty, hospital, and an availability list. Availability is JSON text in doctors, not a separate slots table.

Summary upload

The upload supplies a file and optional patient_id. The source document is processed but not stored; the summary fields and filename are saved.

Not persisted as tables

Staff management shown in the main frontend uses mock data. Model usage, authentication sessions, and dashboard statistics do not have persisted tables in the reviewed code.